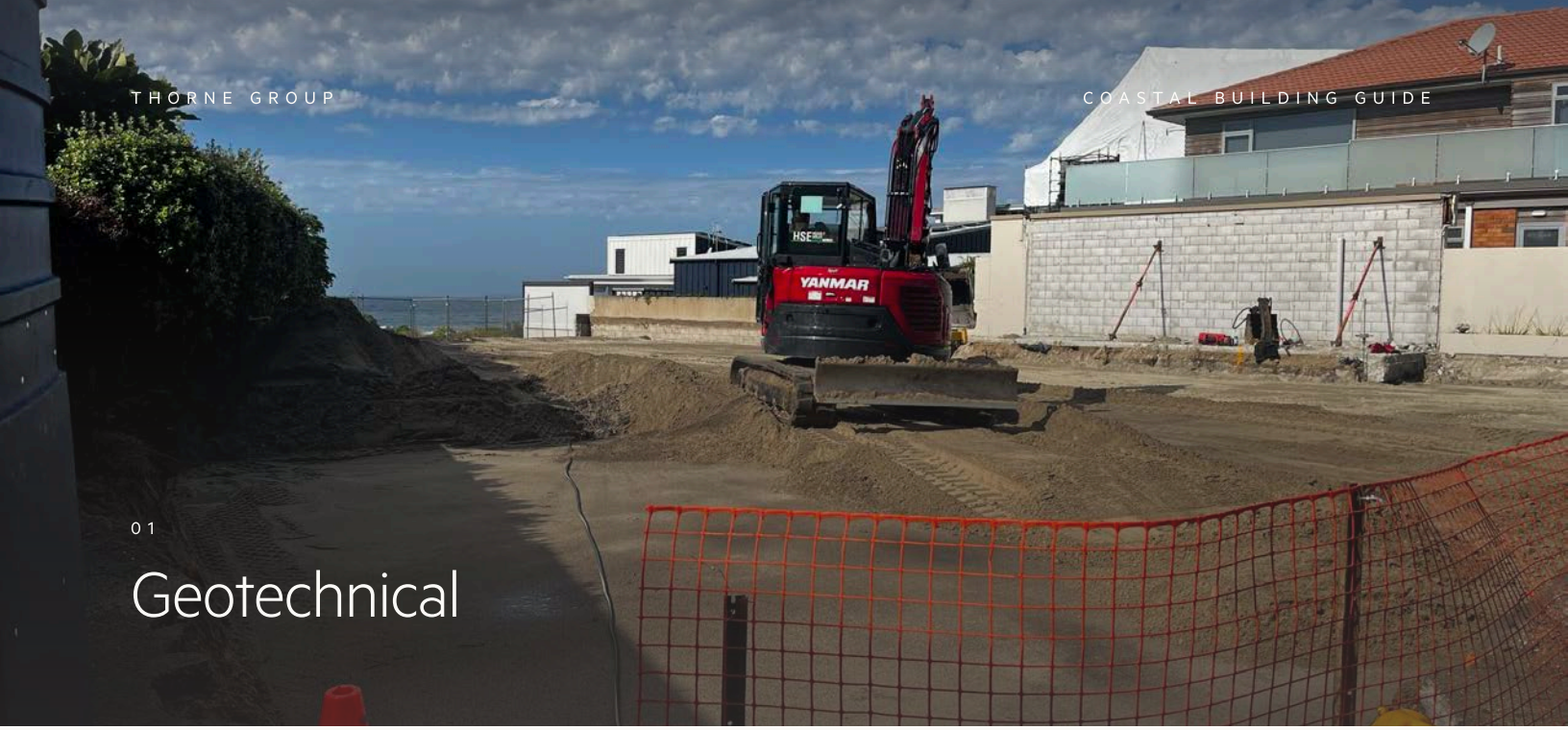
Essential planning guide to **coastal** home building in the Bay of Plenty

Considering building a home within the Bay of Plenty coastal zone or near estuary waterways? Here's what you need to know — a checklist of the unique factors to keep in mind, and how our free Architecture + Build Service can do the leg work for you.



THORNE GROUP



01

Geotechnical

Coastal sites will require a Geotech Investigation Report from an approved council engineer outlining the condition and stability of the ground and reviewing the building area for construction. Coastal foundations need particular attention. The report will include recommendations for the type of foundation to address any liquefaction potential, and recommendations for the structural engineer's design.

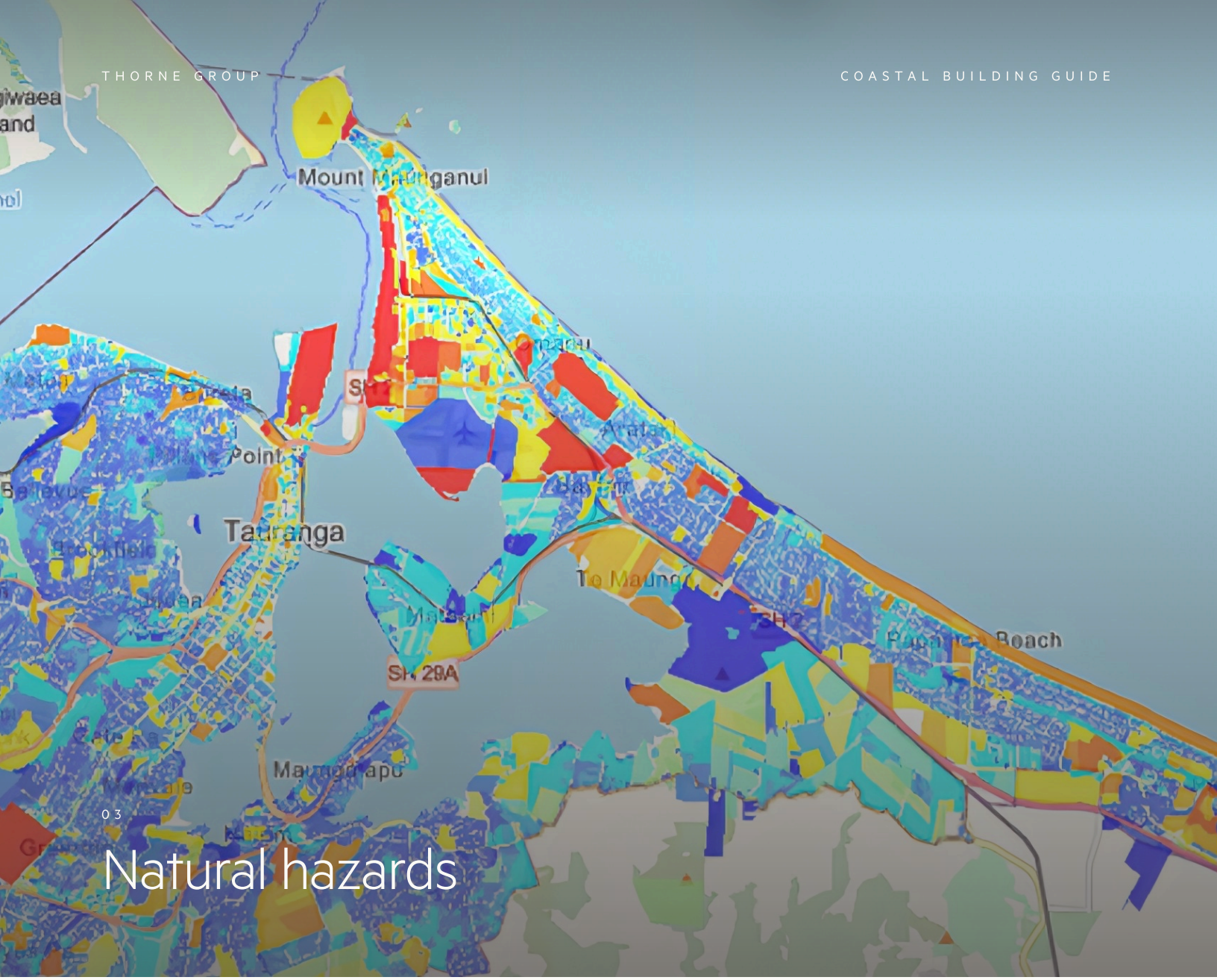
The City or District Plan will set out what is deemed permitted activity, along with any restrictions around boundary setbacks, height, location and site coverage of proposed buildings. If the proposed design does not meet all the criteria, resource consent will be required.



02

Stormwater management

Coastal areas can be prone to weather extremes, and sandy soils may not drain sufficiently quickly during heavy rainfall. Council-provided stormwater catchments are no longer a given, and managing water disbursement is essential to prevent flooding. Most projects will require a site-specific stormwater management system designed by an accredited environmental engineer, taking into account the site coverage, contours, and paved and grassed surfaces. In some instances, a series of interconnected soak holes will provide a suitable solution; however, inground attenuation tanks with slow release are becoming more common. This needs to be considered and factored into your overall project budget.



03

Natural hazards

Council modelling identifies hazard areas that are susceptible to coastal erosion, inundation (flooding from the sea or harbour) and stormwater flooding from rainfall. Scenarios include flood modelling of the likely impacts of a 1% AEP intense rainfall event for both the present day and the year 2130. An independent flood risk assessment may be required to review the council modelling data and determine the effect of the proposed construction on overland flow paths. The flood risk assessment report will set out a minimum finished floor level (FFL) and can be used to support resource consent (if required) and building consent applications.

Check the City or District Council maps to see whether the property is located in a Tsunami Evacuation Zone. Tsunami maps will provide guidance on evacuation routes and meeting points.



04

Coastal erosion

Check the District Plan maps to see whether your site is within an identified coastal erosion area. If it is, you will need to apply for resource consent, and limitations will apply to the size, capacity and type of building that can be constructed.



05

Archaeological

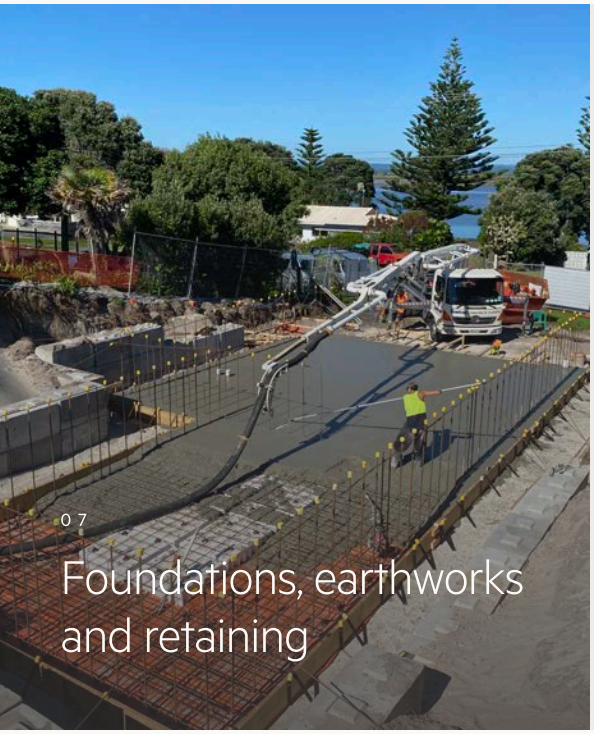
Archaeological sites are common in coastal areas, with 12% of all known archaeological sites in NZ located within 1000m of shorelines. Check the local council (Mapi) website or with Heritage New Zealand whether there is an archaeological site on or near the property. In some instances, an inspection by a qualified archaeologist may be required and an Archaeological Authority recommended. If any items of archaeological significance are identified during excavation, the process to assess and remove them will require input from an archaeologist specialist.



06

Site survey and daylighting survey

If not already undertaken, a site survey will be required. This will set out the boundaries and contours of the site, along with the location of any onsite services and connections to council services. A title search will identify any easements through the property that could impact the design of a new building on the site. Depending on the proposed building design and proximity to neighbouring properties, a daylighting survey may be required to check the building height in relation to boundary rules and the effects of any infringement on neighbouring properties.



07

Foundations, earthworks and retaining

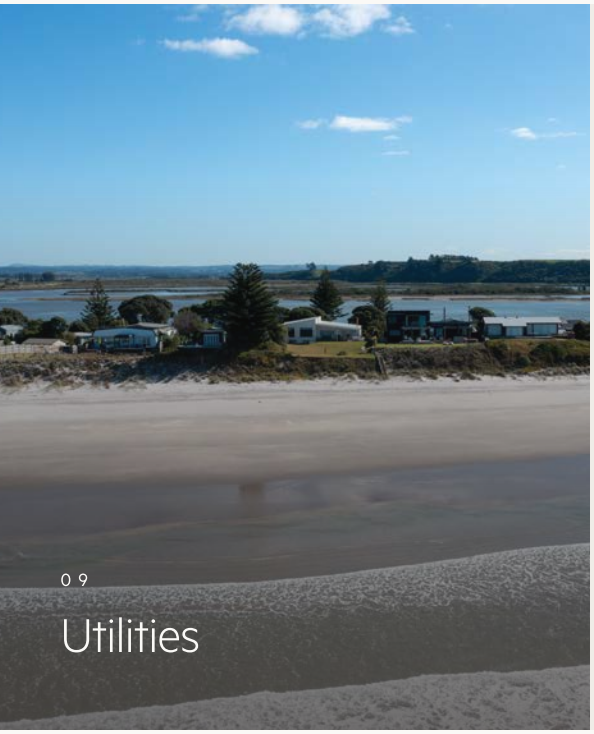
Ground conditions will generally determine the type of foundations suitable for the site. All foundations are required to be Specific Engineer Designed. Sandy soils may require ground improvements, including earthworks to excavate, remove and replace base material with imported fill, recompacted to form a suitable building foundation. Depending on the contours of the site, achieving the ideal finished floor level may require substantial earthworks and/or retaining. If raising the site to take in views is part of your brief, factor this into your overall budget. Additional site investigation, earthworks monitoring and approval from an approved Geotech Engineer will be required before foundation work begins.



08

Wind and exposure

Coastal winds can be strong and unpredictable. All areas in NZ are allocated a wind zone, from Low to Extra High, depending on the level of exposure. Wind direction, speed and frequency will influence the building design, including bracing requirements, roof and wall cladding selection, weathertightness detailing, building entry locations, window size and placement, and provision of shelter for outdoor spaces.



09

Utilities

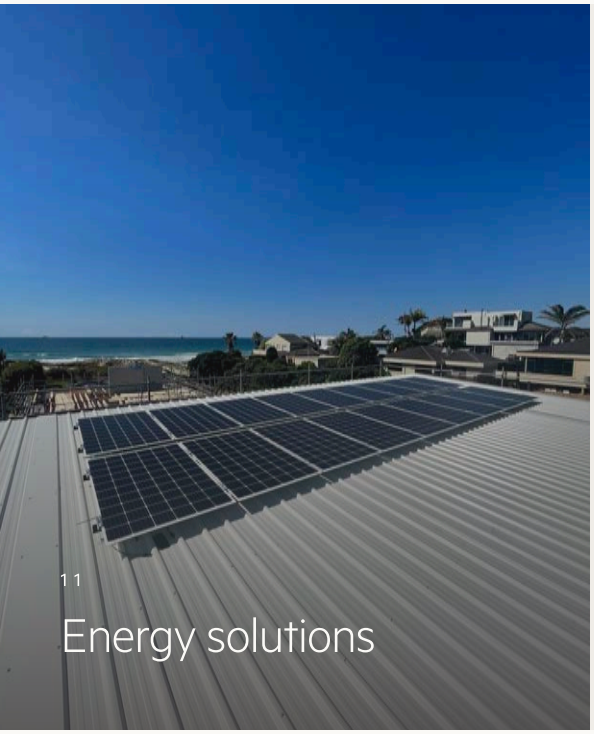
In most instances, supply of power and telecomms to the property boundary is a given. However, where a property has been subdivided or is infill housing, there can be unexpected cost if there is insufficient capacity within the power network for a new connection. Check with the relevant network contractor for any additional costs to upgrade or connect to the power network.



10

Water supply

While most coastal sites will have access to council-supplied water, summer drought conditions may mean water restrictions are imposed during the hottest times of year. You may like to consider installing a rainwater and/or greywater harvesting and storage system to provide non-potable water for garden and lawn use. Alternatively, and particularly if you are considering installing a pool, a bore and pump may be worth considering. If you are installing a pool and connecting to council-supplied water, you will be required to install a backflow prevention device (RPZ) at the property.



11

Energy solutions

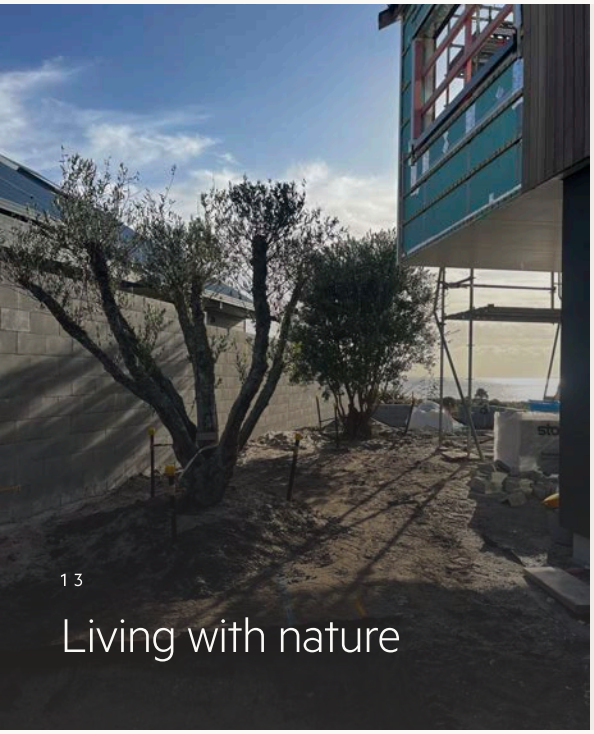
With the cost of electricity continuing to rise, more people are turning to solar power for self-sufficiency and to help manage energy costs. Solar systems are a significant investment, and in most cases battery storage is necessary to gain the most benefit. A well-designed system will take into account the number of occupants and their likely energy requirements for lighting, hot water, heating and cooling, appliances and EV charging. If you are installing a pool, heating it to extend the swimming season is another consideration. While solar can be retrofitted, installing it as part of the build is cheaper and more efficient.



12

Insulation and heating

Comfort is key and, particularly in coastal environments, a cool home is often more important than a warm one. Good design is critical: take advantage of north-facing winter sun for passive solar gain, with wide eaves to provide shade in summer when the sun is at its highest and most intense. Modern insulation, along with thermally broken window joinery and glass, contributes to a home that provides year-round comfort. Factoring in an energy-efficient heating and cooling system will ensure your home is warm in winter and cool in summer, without skyrocketing energy bills.



13

Living with nature

While coastal living is a dream come true for most people, living with the natural elements is not without its challenges. Wildlife, particularly seagulls, may seem intent on making your life difficult. A gable roof ridge is an attractive spot for seagulls to congregate, and guano will quickly spoil the look of a dark roof. Wires and other deterrents may be necessary. Landscaping to suit coastal conditions is also a consideration. Seek advice on coastal planting, particularly around dune areas, where native planting can help stabilise soil and reduce erosion.



14

Outdoor **living**

The point of owning a coastal property is to spend as much time enjoying the outdoors as the seasons will allow. Consider factoring one or two sheltered areas into your design, with roof cover over at least one of them. If your property is beachfront, consider privacy and security both at night and during the day. While features like shutters, sliding screens, retractable blinds and louvres require some ongoing maintenance, they can provide privacy and extend the use of outdoor spaces.

Increase year-round usability with features like an outdoor fire, outdoor kitchen and cooking facilities (BBQ or pizza oven), heating, lighting and built-in seating. An outdoor shower is a must-have for that post-swim wash-off, ideally located near a bathroom with external access. Coastal homes also require carefully considered flooring that is easy to clean and will tolerate sand being tracked inside.





15

Durability and aesthetics

Coastal homes need to be both durable and attractive. Choosing materials that maintain their good looks and can withstand exposure to sea spray and harsh weather is critical for the longevity of your home. Exposed steel, such as structural beams, screens, flashings, railings and balustrades, needs to be appropriately treated and coated. Harsh UV light will accelerate fading and, particularly with exterior materials, may require additional maintenance. Consider low-maintenance cladding and a colour palette that is less susceptible to fading. If selecting materials that require periodic repainting or staining, consider access and factor scaffolding into ongoing maintenance costs.

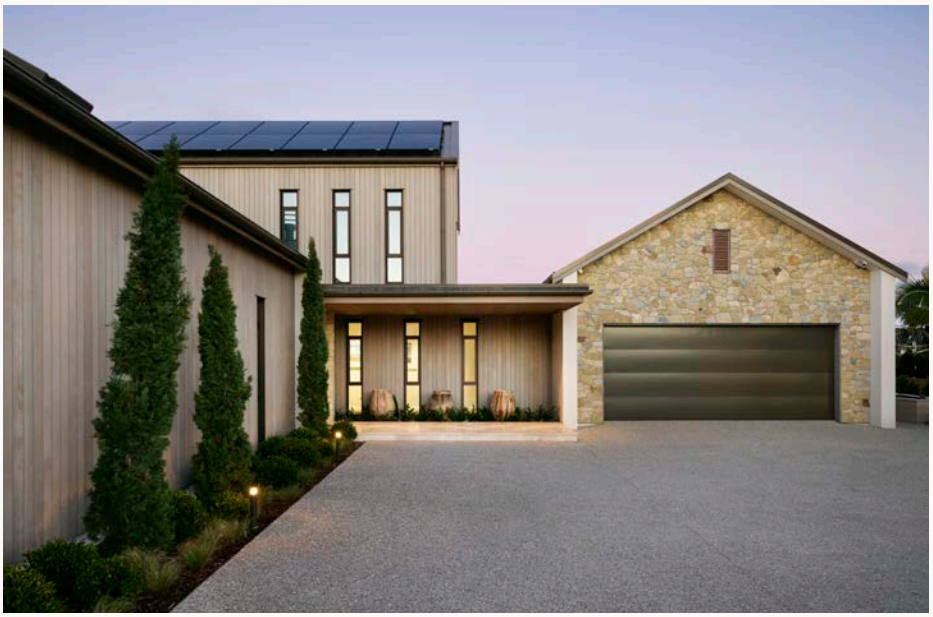
INSPIRATION

Coastal homes we have **built**.

Complex sites across the Bay of Plenty. Each one designed for its land, its climate and the people who live there.



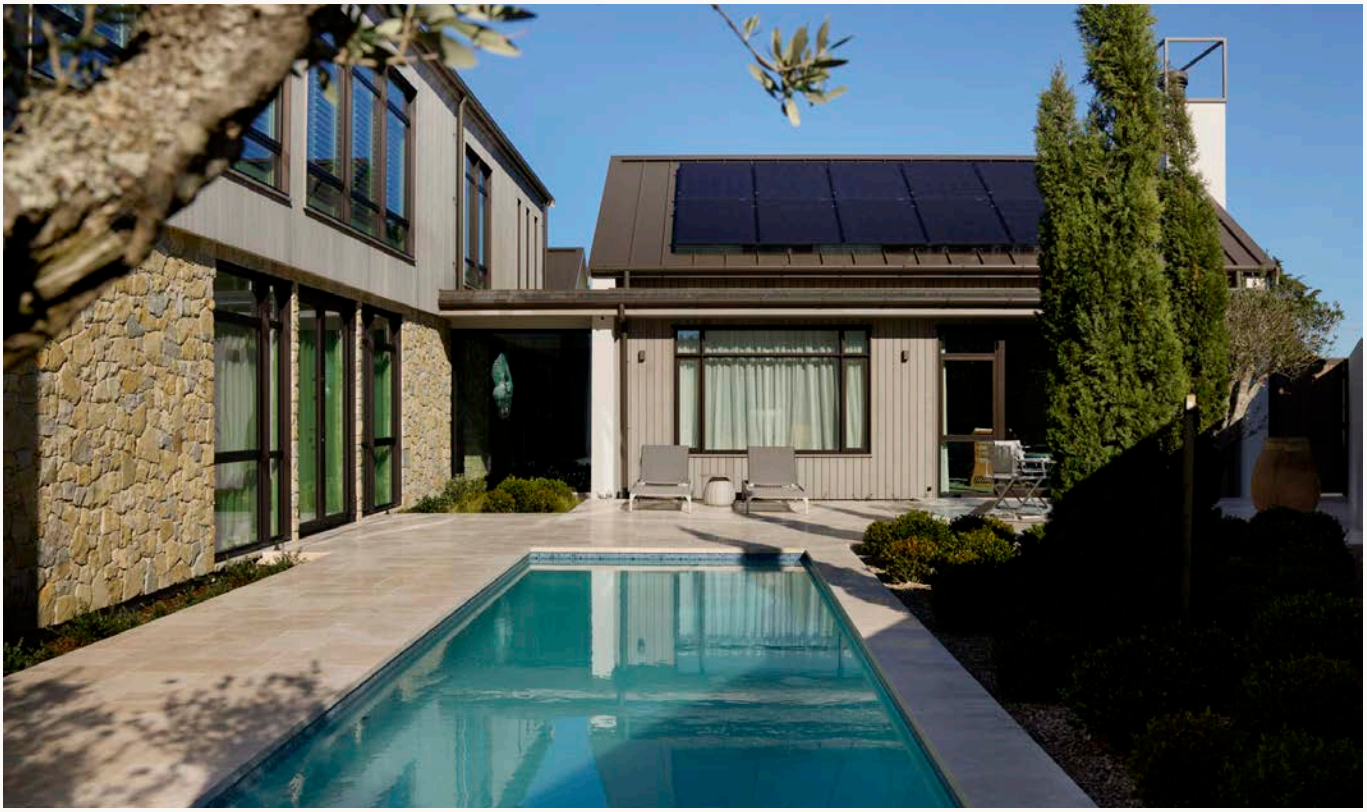
WATERWAYS WHITE HOUSE



COASTAL SANCTUARY



COASTAL SANCTUARY



COASTAL SANCTUARY



ON THE DUNES



ON THE DUNES



MID-CENTURY



MID-CENTURY



MOUNT GOLF COURSE HOME



GOLF RETREAT

The earlier we talk, the more **useful** we can be.

There are numerous considerations with planning a coastal build. If you would like help mapping out next steps, enquire about our Architecture + Build Service and free site assessment. We will talk you through any obstacles or hazards, consents and costs, and recommend the most efficient way to build a coastal home that will last.

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